

ANALYSIS OF A SLIPPAGE TEST FOR THE CHISQUARED DISTRIBUTION

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Introduction

Let $x_1, x_2, \dots, x_n$ be a random sample of size n and suppose one is concerned to test the hypothesis

$$H_0: \text{all } x_j \sim \sigma^2 \chi_m^2$$

where m is a known constant and σ is an unknown constant. Such a set of x_j might arise from the "within cells" sums of squared deviations in an analysis of variance with an equal number of replicates in each cell.

One possible alternative hypothesis to H_0 is that of slippage to the right:

$$\begin{aligned} H_1: x_1 &\sim \lambda^2 \sigma^2 \chi_m^2 \\ x_j &\sim \sigma^2 \chi_m^2 \quad \text{all } j \neq 1 \end{aligned}$$

where $\lambda > 1$ is some unknown constant.

A test procedure for H_0 against H_1 proposed by Cochran (1941) is:

$$\text{Compute } S = \max_{k=1}^n \frac{x_k}{\sum_{k=1}^n x_k}$$

if $S > S_0$ reject H_0 and classify the largest order statistic of the sample as having come from the $\lambda^2 \sigma^2 \chi_m^2$ population.

It has been shown by Truax (1953) that this test procedure is optimal in the sense of maximising the probability of a correct classification subject to reasonable restrictions. Cochran (1941) and Doornbos (1956) showed how Bonferroni's inequalities might be used to find approximations to the fractiles of the distribution of S , but did not explicitly derive the cumulative distribution function. This function and two measures of the performance of the test will be derived in this paper.

Notation

Define

$$z = \sum_{j=1}^n x_j$$

$$z_i = \sum_{\substack{j=1 \\ j \neq i}}^n x_j$$

$$\text{i.e. } z_i = z - x_i$$

$$s_i = \frac{x_i}{z}$$

Let $F_n(\cdot)$ denote the cumulative distribution function of S .

Derivation of the distribution of S under H_0

It is known (e.g. Wilks (1962)) that the joint distribution of any $n-1$ of the size s_i , say $s_1, s_2, \dots, s_{n-1}$ is the Dirichlet distribution with probability density

$$f(s_1 s_2 \dots s_{n-1}) = \frac{\Gamma(\frac{mn}{2})}{[\Gamma(\frac{m}{2})]^n} \prod_{i=1}^{n-1} s_i^{\frac{m}{2}-1} \left(1 - \sum_{i=1}^{n-1} s_i\right)^{\frac{m}{2}-1}$$

in the simplex $\{(s_1, s_2, \dots, s_{n-1}) : s_i \geq 0 \text{ for all } i\}$

$$\text{and } \sum_{i=1}^{n-1} s_i \leq 1 \}$$

$= 0$ elsewhere

$$\text{Noting that } \sum_{i=1}^n s_i = 1, \text{ or } 1 - \sum_{i=1}^{n-1} s_i = s_n$$

we may say for notational simplicity that the joint density for all n of the s_i is

$$(1) \quad f(s_1, s_2 \dots s_n) = \frac{\Gamma(\frac{mn}{2})}{[\Gamma(\frac{m}{2})]^n} \prod_{i=1}^n s_i^{\frac{m}{2}-1}$$

in the positive region of the hyperplane

$$\sum_{i=1}^n s_i = 1'$$

= 0 elsewhere

The marginal distribution of any single s_i is the beta type 1 distribution with probability density

$$(2) \quad g(s_i) = \frac{\Gamma(\frac{mn}{2})}{\Gamma(\frac{m}{2}) \Gamma(\frac{m}{2} [n-1])} s_i^{\frac{m}{2}-1} (1-s_i)^{\frac{m(n-1)}{2}-1}$$

An important property of the Dirichlet distribution is that the joint distribution of the s_i is independent of that of z . This property, well known in the case $n = 2$, is easily demonstrated by transformation from $x_1, x_2, \dots, x_n$ to $s_1, s_2, \dots, s_{n-1}, z$.

Let s be an arbitrarily preassigned value, $0 < s < 1$, and ds a small increment in s .

Since the joint distribution (1) is symmetrical in the s_j , the density of $S = \max s_j$ may be derived as follows

$$(3) \quad P[s < S < s + ds] = \sum_{i=1}^n P(s < s_i < s + ds | \text{all } s_j \leq s_i, j \neq i) \cdot P(\text{all } s_j \leq s_i, j \neq i)$$

$$(4) \quad = \sum_{i=1}^n P(s < s_i < s + ds) P(\text{all } s_j \leq s_i | s < s_i < s + ds).$$

Furthermore

$$(5) \quad P(s < s_i < s + ds) = g(s) ds + o(ds).$$

In analysing the second numerator probability, we note that

$$\begin{aligned} s_j &= \frac{x_j}{z} \\ \frac{1}{s_j} &= \frac{z_j + x_j}{x_j} \\ &= \frac{z_j}{x_j} + 1 \end{aligned}$$

for each $j \neq i$, define

$$(6) \quad v_j = \frac{x_j}{z_i}$$

then $\frac{1}{s_j} = \frac{1}{v_j} \cdot \frac{z_j}{z_i} + 1$

Now the following sets of inequalities are equivalent:

$$(7) \quad s_j \leq s_i$$

$$\Leftrightarrow x_j \leq x_i$$

$$\Leftrightarrow \frac{x_j}{z_i} \leq \frac{x_i}{z_i}$$

$$\Leftrightarrow v_j \leq \frac{s_i}{1-s_i}$$

since $\frac{x_i}{z_i} = \frac{s_i}{1-s_i}$

Next, note that the probability

$$P[s_j \leq s_i \text{ all } j \neq i \mid s < s_i < s + ds]$$

$$(8) \quad = P[s_j \leq s \text{ all } j \neq i \mid s < s_i < s + ds] + O(ds)$$

and the event $s_j \leq s$ all $j \neq i$, is, under the condition $s = s_i$, equivalent to $\max v_j \leq \frac{s}{1-s}$ by (7).

Now $\max v_j$ is a statistic of the same type as S based on a sample of size $n-1$. Further, as noted after equation (2), the distribution of the set of v_j is independent of z_i . As the v_j do not involve x_i , the v_j are thus independent of s_i .

Hence (8) simplifies to

$$P[\max v_j < \frac{s}{1-s}] + O(ds)$$

$$(9) \quad = F_{n-1}\{s/(1-s)\} + O(ds)$$

Thus finally (3) becomes $\sum_{k=1}^n \{g(s)ds + o(ds)\} [F_{n-1}\{s/(1-s)\} + O(ds)]$

$$(10) \quad = n g(s) F_{n-1} [s / (1-s)] ds + 0(ds).$$

Dividing by ds and letting $ds \rightarrow 0$, we obtain the probability density of S as

$$(11) \quad n g(s) F_{n-1} \{s/(1-s)\}$$

One consequence of (11) emerges. It is clear that S cannot exceed 1 in value. Thus $F_n(s) = 1 \Rightarrow 1$ and it follows that if $s > 1/2$, the density of S is $ng(s)$, a result implicit in the work using Bonferroni's inequalities.

The cumulative distribution function of S follows from integration of (11):

$$(12) \quad F_n(s) = n \int_0^s g(t) F_{n-1} \left(\frac{t}{1-t} \right) dt$$

$$\text{Since } F_1(s) = \begin{cases} 0 & s < 1 \\ 1 & s \geq 1 \end{cases}$$

the functions $F_n(s)$ may be found recursively.

The Power of the test

Suppose now that H_1 is true, x_1 being a contaminant from the $\lambda^2 \sigma^2 \chi_m^2$ population, and let S_0 be the fractile used in the test procedure.

As has been pointed out elsewhere (for example David and Paulson (1965)), there are several measures of performance of a slippage test. Here we will derive expressions for two of them:

$$(13) \quad \beta_1 = P[s_1 > S_0]$$

$$\beta_2 = P[s_1 > S_0 \text{ and } (s_j \leq s_1 \text{ all } j \neq 1)]$$

As the test procedure classifies the largest order statistic of the sample as a contaminant if S is significantly large, we see that β_2 is the probability that a contaminant will be correctly identified as such, β_1 , a commonly used approximation to β_2 , does not have a simple sample interpretation such as this.

$$\text{Let } \frac{z_1}{x_1} = t$$

$$\text{then } \frac{1}{s_1} = \frac{z_1 + x_1}{x_1}$$

$$= t + 1$$

Hence $s_1 \geq s$ is equivalent to $t \leq \frac{1}{s} - 1$

Since z_1 is a $\sigma^2 \chi^2_{m(n-1)}$ variate, and is distributed independently of x_1 , which is $\lambda^2 \sigma^2 \chi^2_m$, it follows that $\lambda^2 t$ has a beta type 2 distribution. Let $g^*(.)$ denote the probability density of t .

The first measure β_1 follows at once.

$$\begin{aligned}
 B_1 &= P[s_1 \geq S_0] \\
 &= P\left[t \leq \frac{1}{S_0} - 1\right] \\
 (14) \quad &= \int_0^{\frac{1}{S_0} - 1} g^*(y) dy
 \end{aligned}$$

To derive β_2 let y be a preassigned value and dy a small increment in y . Consider

$$(15) \quad P[(y < t < y + dy) \text{ and } (s_j \geq s_1 \text{ all } j \neq 1)]$$

Defining $v_j = \frac{x_j}{z_1}$ all $j \neq 1$, we can rewrite this as

$$P[(y < t < y + dy) \text{ and } (v_j \leq \frac{1}{t} \cdot \frac{s_1}{1-s_1} \text{ all } j \neq 1)]$$

$$(16) \quad P[(y < t < y + dy) \text{ and } (v_j \leq \frac{1}{t} \text{ all } j \neq 1)]$$

following the arguments leading to (10), we finally arrive at:

$$(17) \quad g^*(y) F_{n-1}(1/y) dy + o(dy)$$

and hence

$$(18) \quad \beta_2 = \int_0^{\frac{1}{S_0} - 1} g^*(y) F_{n-1}\left(\frac{1}{y}\right) dy$$

Comparing (18) with (14), we see that β_1 and β_2 are identical if, over the range of integration, $F_{n-1}(\cdot)$ is 1.

This is equivalent to $S_0 \geq \frac{1}{2}$.

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